Course program and reading list

Semester 1 Year 2023

School: Efi Arazi School of Computer Science M.Sc.

Recommendation Systems

Lecturers:

Dr. Asnat Messica Asnat.Messica@post.runi.ac.il

Dr. Shay Ben-Elazar shay.ben-elazar@post.runi.ac.il

Teaching Assistants:

Mr. Sahar Millis sahar.millis@post.runi.ac.il

Course No.:	Course Type :	Weekly Hours :	Credit:
3639	Elective	3	3

Course Requirements :	Group Code :	Language:
Final Paper	231363901	Hebrew

Prerequisites

Prerequisite:

52 - Calculus I
53 - Calculus II
54 - Linear Algebra I
55 - Linear Algebra II
56 - Discrete Mathematics
59 - Data Structures
69 - Logic and Set Theory
417 - Introduction to Computer Science
3141 - Machine Learning from Data
3598 - Deep Learning and Image Understanding **OR** 3600 - Deep Learning

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Course Description

Recommender Systems

Syllabus

Lecturer: Dr. Asi Messica, Dr. Shay Ben Elazar

Teaching assistant: Mr. Sahar Millis

Introduction

Recommender systems are widely used by online companies worldwide to provide the most relevant content to their users. They changed how users find relevant items in various areas such as e-commerce, content websites, social media, and more.

This course aims to provide the students with both theoretical and practical aspects of recommender systems. We will study machine learning algorithms for personalized recommendations, including recent algorithms, advanced evaluation, research topics, and implementation tools.

Meeting plan*

Meeting	Topic	Assignment
1	Introduction to recommender systems	
2 - 3	Collaborative filtering and recommender systems evaluation	Exercise #1
4 - 5	Matrix factorization	
6 - 7	Content-based recommender systems and explainability in recommender systems	Exercise #2
8	Hybrid approaches	
9 - 11	Deep learning for recommender systems and guest lectures	Exercise #3
12	Final project presentations	
13	Summary	

* A tentative outline. Participation in guest lectures and students' final project presentations is mandatory.

Course Goals

Course Goals:

This course aims to provide the students with both theoretical and practical aspects of recommender systems. We will study machine learning algorithms for personalized recommendations, including recent algorithms, advanced evaluation, research topics, and implementation tools.

Grading

Assessment

Final project – 40%

Exercises – 60%

Exercise #1: Matrix factorization implementation and evaluation (25%)

Exercise #2: Content-based recommendations (15%)

Exercise #3: Deep learning for recommender systems and hybrid approach (20%)

Final Project:

The goal of the final project is to expose the students to advanced topics in recommender systems that were not covered in class. The students will select an innovative research paper, present it, implement it and add an innovative enhancement.

Final project submissions will be in pairs.

Learning Outcomes

Learning Outcomes:

Graduates of the course will be ready to join a personalization or recommendation team in the industry and participate in advanced research in this area.

Lecturer Office Hours

Thursday 21 - 22

Additional Notes

Participation in guest lectures and students' final project presentations is mandatory.

Reading List

Bibliography

- Ricci, F., Rokach, L., Shapira, B., Kantor, P.B., 2011. Recommender Systems Handbook.
- Aggarwal, C.C., 2016. Recommender systems (Vol. 1). Cham: Springer International Publishing.
- Koren, Y., Bell, R. and Volinsky, C., 2009. Matrix factorization techniques for recommender systems. Computer, (8), pp.30-37.
- Hu, Y., Koren, Y. and Volinsky, C., 2008, December. Collaborative filtering for implicit feedback datasets. In Data Mining, 2008. ICDM'08. Eighth IEEE International Conference on (pp. 263-272). IEEE
- Rendle, S., 2010, December. Factorization machines. In 2010 IEEE International Conference on Data Mining (pp. 995-1000). IEEE

Additional papers will be provided before every meeting